

Program 1: Student Record Management System

```
import java.util.Scanner;

class Student {
    int studentId;
    String studentName;
    String course;
    double aas, java, python;

    Student() {
        studentId = 0;
        studentName = "Unknown";
        course = "Not Assigned";
        aas = 0;
        java = 0;
        python = 0;
    }

    Student(int id, String name, String c, double a, double j, double p) {
        studentId = id;
        studentName = name;
        course = c;
        aas = a;
        java = j;
        python = p;
    }

    void acceptDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student ID: ");
        studentId = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Student Name: ");
        studentName = sc.nextLine();

        System.out.print("Enter Course: ");
        course = sc.nextLine();

        System.out.print("Enter AAS Marks: ");
        aas = sc.nextDouble();

        System.out.print("Enter JAVA Marks: ");
        java = sc.nextDouble();

        System.out.print("Enter Python Marks: ");
        python = sc.nextDouble();
    }

    double calculateTotal() {
        return aas + java + python;
    }
}
```

```

double calculatePercentage() {
    return calculateTotal() / 3;
}

void displayDetails() {
    System.out.println("Student ID: " + studentId);
    System.out.println("Student Name: " + studentName);
    System.out.println("Course: " + course);
    System.out.println("AAS Marks: " + aas);
    System.out.println("JAVA Marks: " + java);
    System.out.println("Python Marks: " + python);
    System.out.println("Total Marks: " + calculateTotal());
    System.out.println("Percentage: " + calculatePercentage());
}
}

public class StudentRecord {
    public static void main(String[] args) {
        Student s1 = new Student();

        System.out.println("Default Student:");
        s1.displayDetails();

        System.out.println();

        Student s2 = new Student(101, "Rahul Sharma", "BSc CS", 85, 90, 88);

        System.out.println("Parameterized Student:");
        s2.displayDetails();
    }
}

```

Program 2: Bank Account Management System

```

import java.util.Scanner;

class BankAccount {
    long accountNumber;
    String accountHolderName;
    double balance;

    BankAccount() {
        accountNumber = 0;
        accountHolderName = "Unknown";
        balance = 0.0;
    }

    BankAccount(long number, String name, double amount) {
        accountNumber = number;
        accountHolderName = name;
        balance = amount;
    }

    void deposit(double amount) {
        balance = balance + amount;
    }
}

```

```

    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawal successful");
        } else {
            System.out.println("Insufficient balance");
        }
    }

    void displayBalance() {
        System.out.println("Current Balance: " + balance);
    }

    void displayAccountDetails() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder: " + accountHolderName);
        System.out.println("Balance: " + balance);
    }
}

public class BankAccountDemo {
    public static void main(String[] args) {
        BankAccount b1 = new BankAccount();

        System.out.println("Default Account:");
        b1.displayAccountDetails();

        System.out.println();

        BankAccount b2 = new BankAccount(123456789, "Rahul Sharma", 5000);

        b2.displayAccountDetails();

        b2.deposit(2000);
        b2.withdraw(1000);

        System.out.println();
        System.out.println("Updated Account:");
        b2.displayAccountDetails();
        b2.displayBalance();
    }
}

```

Program 3: Bank Account Management System for 5 Customers

```

import java.util.Scanner;

class BankAccount {
    long accountNumber;
    String accountHolderName;
    double balance;
}

```

```

    BankAccount(long number, String name, double amount) {
        accountNumber = number;
        accountHolderName = name;
        balance = amount;
    }

    void deposit(double amount) {
        balance = balance + amount;
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
        } else {
            System.out.println("Insufficient balance");
        }
    }

    void display() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Holder: " + accountHolderName);
        System.out.println("Balance: " + balance);
        System.out.println();
    }
}

public class BankCustomers {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        BankAccount accounts[] = new BankAccount[5];

        for (int i = 0; i < 5; i++) {
            System.out.println("Enter details of customer " + (i + 1));

            System.out.print("Account Number: ");
            long number = sc.nextLong();

            sc.nextLine();

            System.out.print("Account Holder Name: ");
            String name = sc.nextLine();

            System.out.print("Opening Balance: ");
            double balance = sc.nextDouble();

            accounts[i] = new BankAccount(number, name, balance);
            System.out.println();
        }

        accounts[0].deposit(1000);
        accounts[1].withdraw(500);

        System.out.println("Customer Details:");

        for (int i = 0; i < 5; i++) {
            accounts[i].display();
        }
    }
}

```

```

    }
}

```

Program 4: Single Inheritance – Smart Home Automation

```

class Device {
    int deviceId;
    String deviceName;
    boolean powerStatus;

    Device(int id, String name) {
        deviceId = id;
        deviceName = name;
        powerStatus = false;
    }

    void turnOn() {
        powerStatus = true;
    }

    void turnOff() {
        powerStatus = false;
    }
}

class SmartLight extends Device {
    int brightnessLevel;
    String colorMode;

    SmartLight(int id, String name, int brightness, String color) {
        super(id, name);
        brightnessLevel = brightness;
        colorMode = color;
    }

    void changeBrightness(int brightness) {
        brightnessLevel = brightness;
    }

    void display() {
        System.out.println("Device ID: " + deviceId);
        System.out.println("Device Name: " + deviceName);
        System.out.println("Power: " + (powerStatus ? "ON" : "OFF"));
        System.out.println("Brightness: " + brightnessLevel);
        System.out.println("Color Mode: " + colorMode);
    }
}

public class SmartHome {
    public static void main(String[] args) {
        SmartLight light = new SmartLight(101, "Living Room Light", 70,
"Warm");

        light.turnOn();
    }
}

```

```

        light.changeBrightness(90);
        light.display();

        light.turnOff();

        System.out.println();
        light.display();
    }
}

```

Program 5: Multilevel Inheritance – Online Shopping System

```

class User {
    int userId;
    String userName;
    long mobileNumber;

    User(int id, String name, long mobile) {
        userId = id;
        userName = name;
        mobileNumber = mobile;
    }
}

class Customer extends User {
    String address;
    int orderId;

    Customer(int id, String name, long mobile, String address, int order) {
        super(id, name, mobile);
        this.address = address;
        orderId = order;
    }
}

class PremiumCustomer extends Customer {
    String membershipType;
    double discountPercentage;

    PremiumCustomer(int id, String name, long mobile, String address,
                    int order, String membership, double discount) {
        super(id, name, mobile, address, order);
        membershipType = membership;
        discountPercentage = discount;
    }

    double calculateDiscount(double amount) {
        return amount * discountPercentage / 100;
    }

    void displayBill(double amount) {
        double discount = calculateDiscount(amount);
        double finalAmount = amount - discount;
    }
}

```

```

        System.out.println("Order Amount: " + amount);
        System.out.println("Discount: " + discount);
        System.out.println("Final Bill: " + finalAmount);
    }

    void displayDetails() {
        System.out.println("User ID: " + userId);
        System.out.println("User Name: " + userName);
        System.out.println("Mobile Number: " + mobileNumber);
        System.out.println("Address: " + address);
        System.out.println("Order ID: " + orderId);
        System.out.println("Membership: " + membershipType);
        System.out.println("Discount: " + discountPercentage + "%");
    }
}

public class OnlineShopping {
    public static void main(String[] args) {
        PremiumCustomer p = new PremiumCustomer(
            101,
            "Rahul Sharma",
            9876543210L,
            "Mumbai",
            5001,
            "Gold",
            15
        );

        p.displayDetails();

        System.out.println();

        p.displayBill(5000);
    }
}

```

Program 6: Static Keyword – Student Management System

```

class Student {
    int rollNumber;
    String studentName;
    static String collegeName = "Mithibai College";
    static int count = 0;

    Student(int roll, String name) {
        rollNumber = roll;
        studentName = name;
        count++;
    }

    void display() {
        System.out.println("Roll Number: " + rollNumber);
        System.out.println("Student Name: " + studentName);
        System.out.println("College Name: " + collegeName);
        System.out.println();
    }
}

```

```

    }

    static void totalStudents() {
        System.out.println("Total Students: " + count);
    }
}

public class StudentManagement {
    public static void main(String[] args) {
        Student s1 = new Student(1, "Rahul");
        Student s2 = new Student(2, "Aman");
        Student s3 = new Student(3, "Priya");

        s1.display();
        s2.display();
        s3.display();

        Student.totalStudents();
    }
}

```

Program 7: Abstract Class – Digital Payment Gateway

```

abstract class Payment {
    double amount;

    Payment(double amount) {
        this.amount = amount;
    }

    void displayReceipt() {
        System.out.println("Transaction Amount: ₹" + amount);
    }

    abstract void processPayment();
}

class CryptoPayment extends Payment {
    String walletAddress;

    CryptoPayment(double amount, String wallet) {
        super(amount);
        walletAddress = wallet;
    }

    void processPayment() {
        System.out.println("Payment processed successfully");
        System.out.println("Secure blockchain transfer to: " +
walletAddress);
    }
}

public class DigitalPayment {
    public static void main(String[] args) {
        CryptoPayment p = new CryptoPayment(

```

```
        5000,  
        "0xABC123456789"  
    );  
  
    p.displayReceipt();  
    p.processPayment();  
}  
}
```